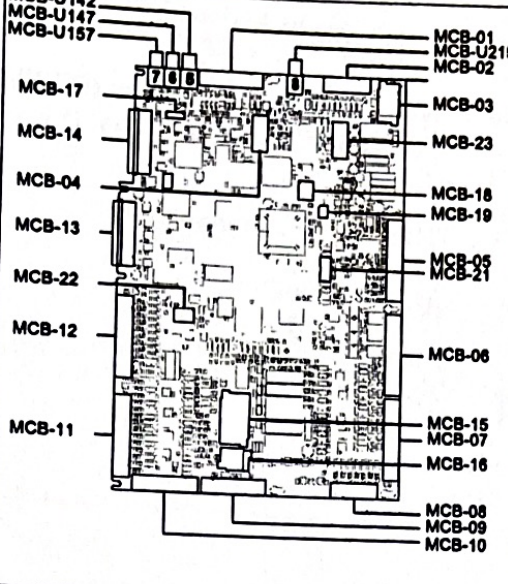
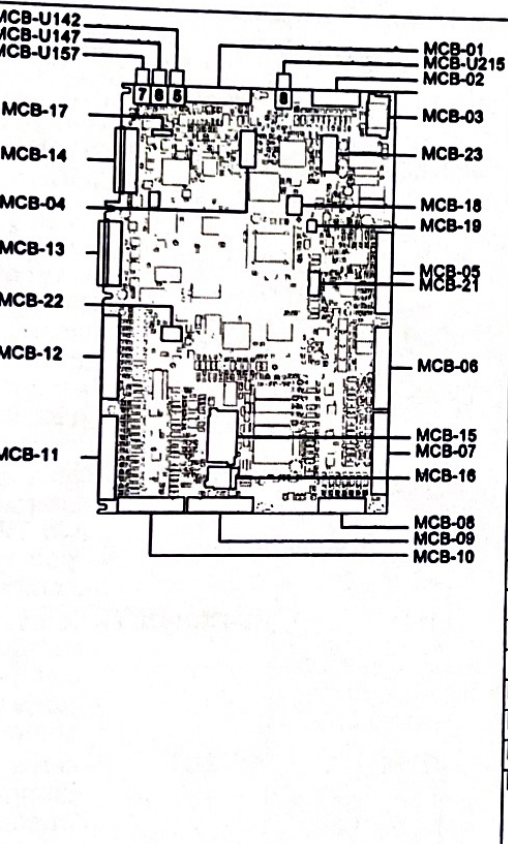
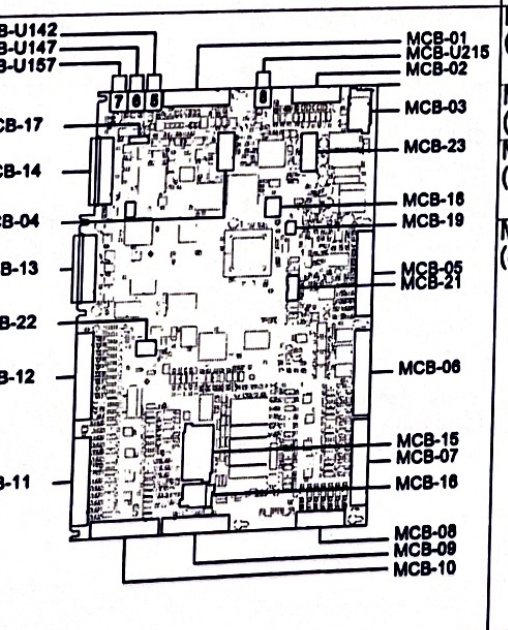


Illustration	Connector	Connections
	MCB-01	- communication with the machine terminal and the TRIAX print - power supply to the machine terminal - Drive Disable signals for the TRIAX and T_TRIAX prints. These signals are passed on to the respective print via the EPDAF print.
	MCB-02	- parallel communication to the weaving mill central computer. - connection to the diagnostic plug.
	MCB-03	- power supply coming from the EPDAF print. - emergency stop signal coming from the EPDAF print.
	MCB-04	resolver signals from the machine control system.
	MCB-05	connection to the warp tension sensors, oil temperature sensor, outside cam motion oil level sensor, control signal for the fuses of the EPDAF print.
	MCB-06	connection to the dobby.
	MCB-07	connection to the push-button detections and jumpers.
	MCB-08	- connection to the high voltage plate thermal detection elements. - connection to the vacuum pump motor bimetal.
	MCB-09	connection to: - the signal lights - the EPDAF print warning lights signal - the electrical safety brake valve - the pickfinder valve
	MCB-10	connection to the hand safety and the push-buttons (protection logic)
	MCB-11	connection to: - the proximity switches - the oil filter - the jacquette - the batching motion
	MCB-12	connection to the warp stop motion, selvage and waste yarn detections and to the filling detector
	MCB-13	directly plugged into IBR-01.
	MCB-14	directly plugged into MIDB/R-01 or MDSB-01
	MCB-15	connection to: - the motor switches on the high voltage plate - the POWER ON button - the oil pressure detection and the EPDAF print emergency stop signal
	MCB-16	power supply coming from the EPDAF print
	MCB-17	for Picanol service
	MCB-18	directly plugged in to DDMP-03
	MCB-19	for Picanol service
	MCB-20	not present on the print
	MCB-21	for Picanol service
	MCB-22	directly plugged in to DDMP-03
	MCB-23	- PICAN pulse train - EVENTLINE - FTEN
	MCB-U142 (5)	optical connection to the T_TRIAX print for the angle position (receiver)
	MCB-U147 (6)	optical connection to the T_TRIAX print for the angle position (transmitter)
	MCB-U157 (7)	optical connection to the T_TRIAX print for the angle position (receiver)
	MCB-U215 (8)	optical connection to the TRIAX print for the angle position (transmitter)

Preliminaries

1. Use the USB Memory Stick to make a backup of the machine-settings. For more information, see "Sw-transfer [71]".
 2. Copy the machine software to a USB Memory Stick.
For more information, see "Sw-transfer [71]".
-  The MCB print may not come into contact with a metal table or any other metal component.

DDMP print

Remove the DDMP print from the old MCB print and plug it in on the new one.

After the replacement

3. Copy the machine software to the machine (MCB print). See "Sw-transfer [71]".
4. Copy the machine settings to the machine (MCB print). See "Sw-transfer [71]".
5. Re-synchronize the machine.

Sw-transfer

Display type	Transfer medium	Further explanation
Interactive display	USB Memory Stick	see the section "The Interactive display", chapter "General: USB Stick"

DDMP print

Name

Direct Drive main Motor Processor

Version

DDMP-4: BE214832

General

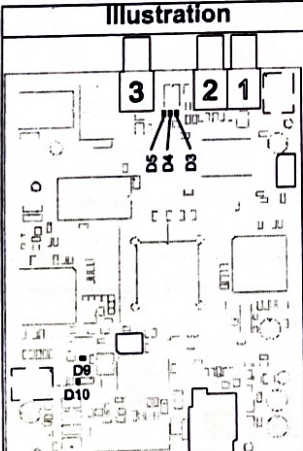
- This print contains a microprocessor using its own software. This software is checked upon:
 - power-on;
 - changing of machine software;
- If the check indicates that the software is not compatible with the MCB print software, new software is automatically loaded into the DDMP print.

Function

This print:

- **provides the protection logic**
The DDMP and MCB prints together provide the protection logic. The protection logic prevents the machine from being controlled in an undesirable manner.
- **controls the main motor**
The DDMP print directly controls the driver stages on the NG_LDEC module via an optical connection.

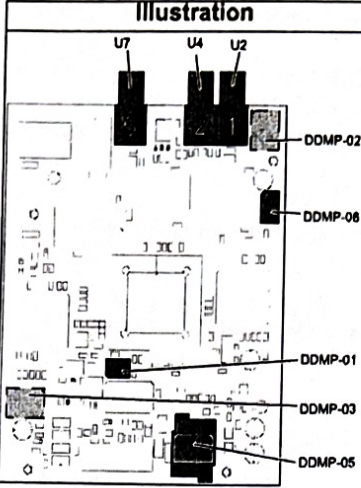
Signals

Illustration	LED	Color	Name	The LED...
	D3	yellow	Data In	flashes in the event of data communication with the NG_LDEC module. This is a check of the optical connection between the DDMP and NG_LDEC module.
	D4		Data Out	
	D5		RPT	
	D9	green	Reset	is lit if the +5,2VDC on this print is more than 5 VDC. If the +5,2 VDC is less than 5VDC, the RESET line is low. The microprocessor halts and will not start again until the +5,2 VDC once more exceeds 5 VDC.
	D10		Watchdog	

Fusing

There are no fuses on the print.

Connections

Illustration	Connector	Connections
	DDMP-01	for Picanol service
	DDMP-02	directly plugged in to the MCB-18
	DDMP-03	directly plugged in to the MCB-22
	DDMP-04	not present on the print
	DDMP-05	- AC-GND - signal from the emergency stops - signals from the K-MM motor switch - Drive Disable for the main motor power control modules
	DDMP-06	for Picanol service
	DDMP-U2 (1)	optical connection to NG_LDEC-U9 (1) (receiver)
	DDMP-U4 (2)	optical connection to NG_LDEC-U6 (2) (transmitter)
	DDMP-U7 (3)	not used

MIDB/R print

Name

Machine IDentification Board/Rapier

Version

MIDB/R-1: BE217401

General



This print cannot be interchanged with a print from another weaving machine.

This print is only used if the machine is NOT equipped with a Board rack [77].If a board rack is present, the MIDSB print [72] is used.

Function

This print constitutes the electronic identification plate of the weaving machine.

This print contains:

- the MID settings: machine configuration;
- the serial number. This number is also indicated on the machine identification plate.

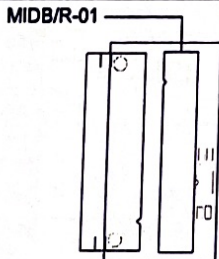
Signals

There are no LEDs on the print.

Fusing

There are no fuses on the print.

Connections

Illustration	Connector	Connections
	MIDB/R-01	directly plugged in to the MCB-14

If problems occur

Contact Picanol.

MIDSB print

Name

Machine IDentification Board with Small Bus print

Version

MIDSB-1: BE312458

General



This print cannot be interchanged with a print from another weaving machine.

This print is only used if the machine is equipped with a Board rack [77].If this is NOT the case, the MIDB/R print [72] is used.

Function

This print constitutes the electronic identification plate of the weaving machine.

This print contains:

- the MID settings: machine configuration;
- the serial number. This number is also indicated on the machine identification plate.

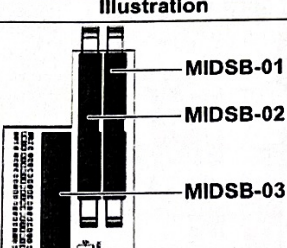
Signals

There are no LEDs on the print.

Fusing

There are no fuses on the print.

Connections

Illustration	Connector	Connections
	MIDSB-01	connection with SBPO-01
	MIDSB-02	connection with SBPO-02
	MIDSB-03	directly plugged in to the MCB-14

If problems occur

Contact Picanol.

IBR print

Name

Insertion Board Rapier

Version

IBR/8C-7: BE222400

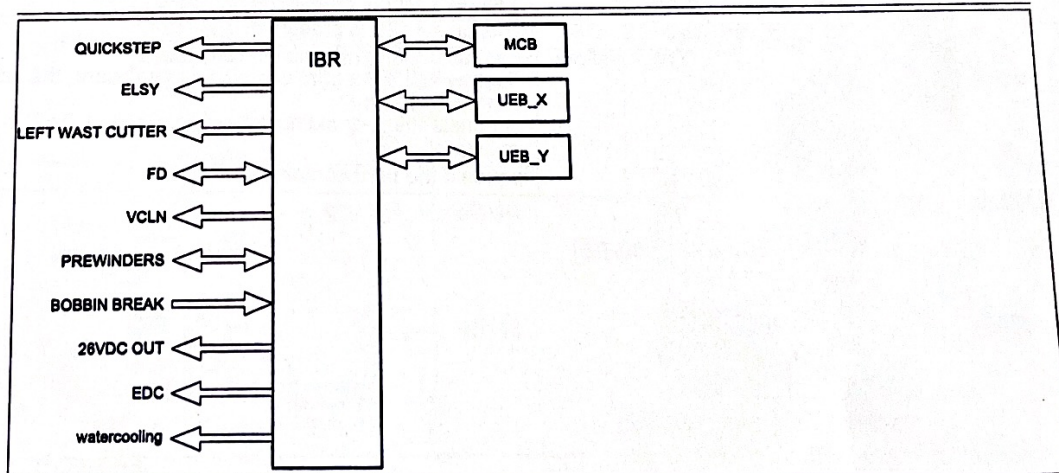
IBR/12C-7: 222402



Both prints have the same functions.

- The IBR/8C-7 is equipped to control maximally 8 channels,
- the IBR/12C-7 is equipped to control maximally 12 channels.

Function

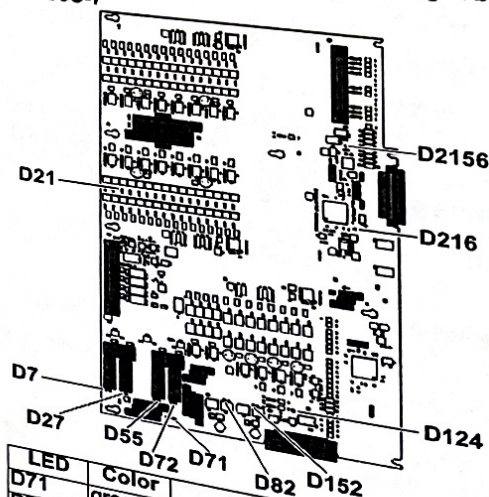




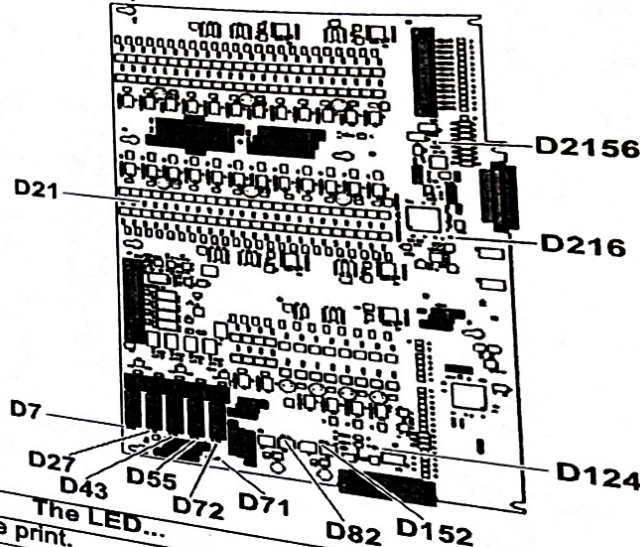
Every LED color has a specific meaning:

- Green LED: Indication.
- Yellow LED: danger, don't touch (i.e.: at high voltages)
- Red LED: Indicates a problem (e.g. a blown fuse)

IBR/8C-7

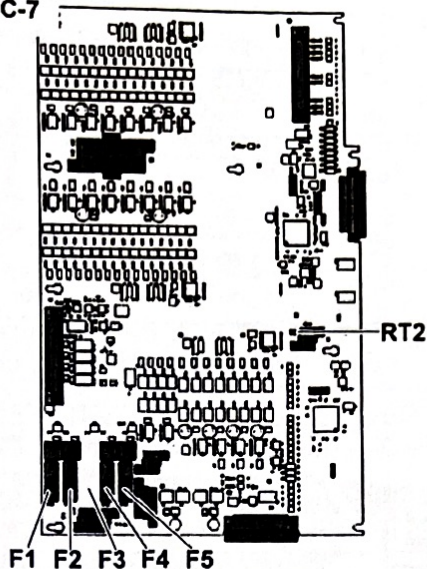


IBR/12C-7

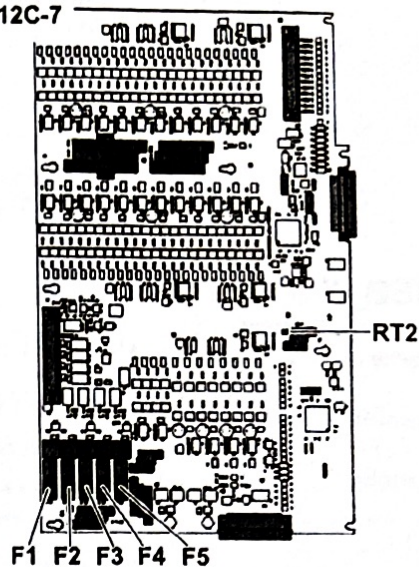


LED	Color	The LED...
D71	green	lights up if +26V is present on the print.
D7	green	lights up if the corresponding fuse F1 fails.
D27	red	lights up if the corresponding fuse F2 fails.
D43		lights up if the corresponding fuse F3 fails. (only with IBR/12C-7)
D55		lights up if the corresponding fuse F4 fails.
D72		lights up if the corresponding fuse F5 fails.
D21		lights up if too much power is going to the bobbin break sensor.
D152	green	lights up if +3,3V is present on the print.
D124	green	lights up if +5V is present on the print.
D82	yellow	lights up if +12V is present on the print.
D215	yellow	flashes if the print contains software, the software is automatically reprogrammed;
		is constantly lit or extinguished at upload of the PROM or in case of invalid software.
D216		flashes if the print contains software.

IBR/8C-7



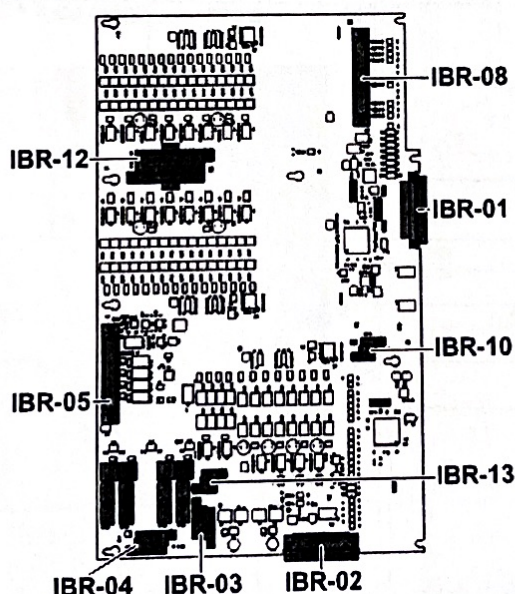
IBR/12C-7



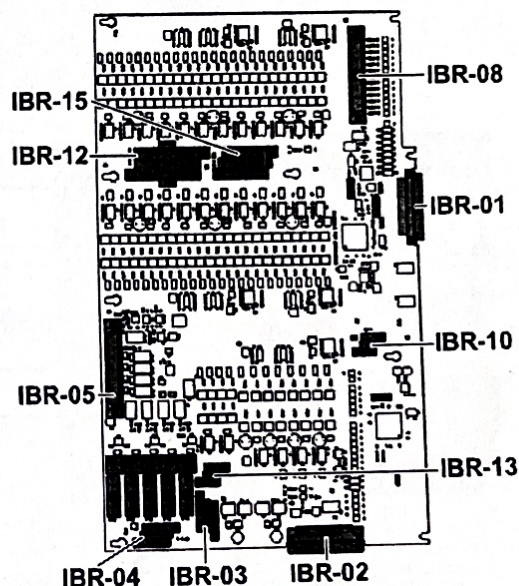
Fuse	Corresponding coil	Circuit	Value	Order Number
F1	L1	+26VQS5_8 (QuickStep color 5 up to and including 8)	8 AT	N1020091
F2	L2	+26VQS1_4 (QuickStep color 1 up to and including 4)		
F3	- B1.	+26VQS9_12 (QuickStep color 9 up to and including 12)		
F4	L4	+26VELSY		
F5	L5	+26VP5C (IBR-05)		
RT2	—	BOBRK (short circuit protection)	—	—

Connections

IBR/8C-7



IBR/12C-7



Connector	Connections
IBR-01	Connected to the MCB print.
IBR-02	Connected to the UEB-X print.
IBR-03	Connected to the right-hand ELSY module and its push button, and to the left-hand waste cutter.
IBR-04	Power supply coming from the PDAF print.
IBR-05	Connected to the filling detectors (FD1 and FD2), the gripper and the gripper clamp cleaning valve, EDC and water cooling.
IBR-06	Not present on the print.
IBR-07	
IBR-08	Connected to the prewinders and the bobbin break sensors.
IBR-09	For Picanol service.
IBR-10	Power supply coming from the PDAF print.

Connector	Connections
IBR-11	Not present on the print.
IBR-12	Connected to the filling presenter motors. (color 1 up to and including 8)
IBR-13	Connected to the left-hand ELSY module and its push button.
IBR-14	Not present on the print.
IBR-15	Connected to the filling presenter motors. (color 9 up to and including 12)
IBR-16 up to and including IBR-20	Not present on the print.

UEB_X and UEB_Y

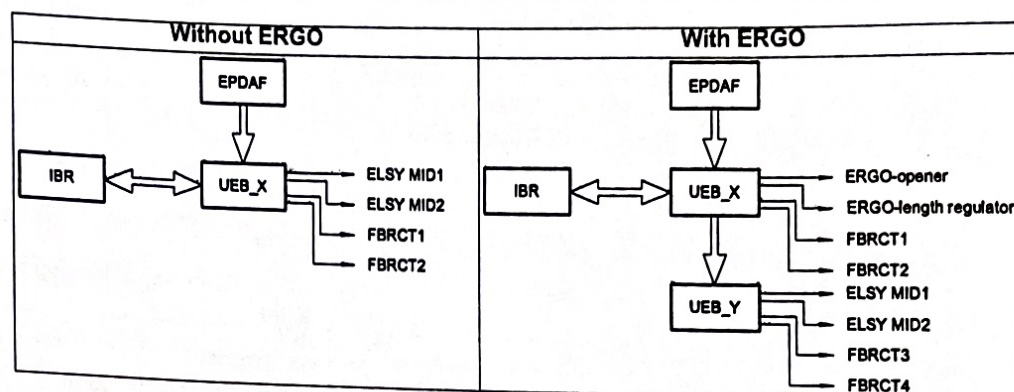
Name

Universal Extension Board

Version

UEB-5: BE222766

Function



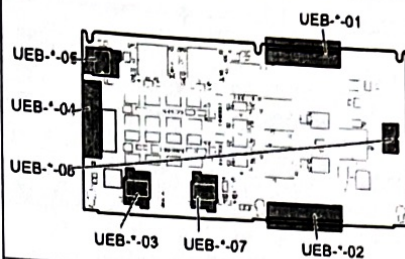
Signals

Illustration	LED	Color	The LED...
	D1	red	lights up if the corresponding fuse F1 fails.
	D3	green	lights up if +26V is present on the print.

Fusing

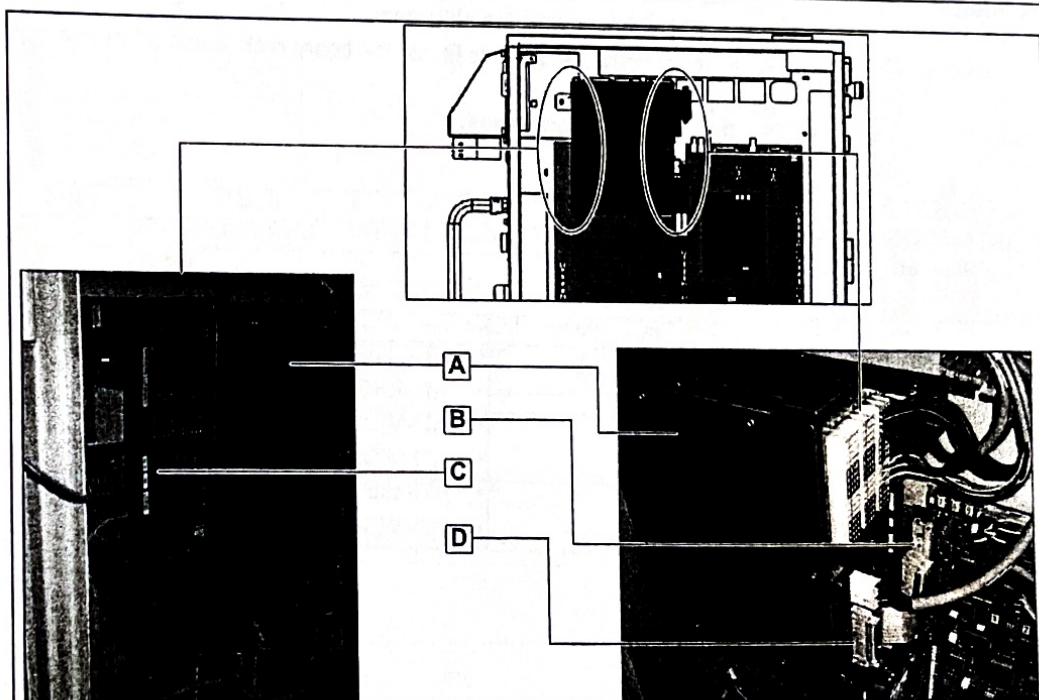
Illustration	Fuse	Corresponding coil	Circuit	Value	Order Number
	F1	L1	+26VM1_2	8 AT	N1020091

Illustration



Connector	UEB_X (with ERGO)	UEB_X (without ERGO)	UEB_Y
UEB-*01	Directly plugged into IBR-02.		Directly plugged into UEB_X-02.
UEB-*02	Directly plugged into UEB_Y-01.		
UEB-*03	Connected to the ERGO opener.	Connected to the left middle ELSY.	Not used
UEB-*04	Connected to the 2 fabric cutters.		
UEB-*05	Power supply coming from the EPDAF print.		
UEB-*06	Not present on the print.		
UEB-*07	Connected to the ERGO length controller.	Connected to the right middle ELSY.	
UEB-*08	For Picanol service.		

Board rack



- A. Board rack [77]
- B. MDSB print [72]
- C. Board Rack > ATB X, ATB Y and ATB Z print [78]
- D. Board rack > SBPO-print [77]

Board rack > SBPO-print

Version

SBPO-1: BE222472

Function

This print devices the following items to the prints that are plugged into the SBPO print:

- bus signals originating from the MCB print;
- power supplied from the EPDAF print.

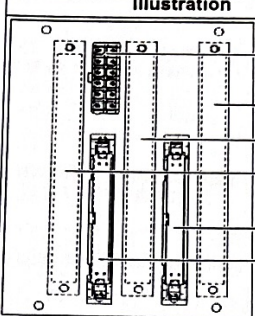
Signals

There are no LEDs on the print.

Fusing

There are no fuses on the print.

Connections

Illustration	Connector	Connections
	SBPO-01	connection with MIDSB-01
	SBPO-02	connection with MIDSB-02
	SBPO-03	power supply coming from the EPDAF print
	SBPO-04	directly plugged in to ATB_X-02
	SBPO-05	directly plugged in to ATB_Y-02
	SBPO-06	directly plugged in to ATB_Z-02

Board Rack > ATB_X, ATB_Y and ATB_Z print

Name

Air Tucker Board

Version

ATB-2: BE312104

General

The print:

- This print controls the airtuckers.
- Maximally 3 ATB prints fit into the board rack, these prints can control maximally 6 airtuckers.

One airtucker comprises:

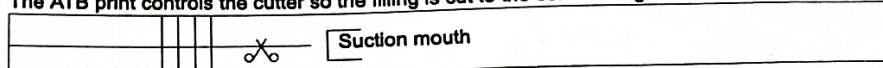
- a cutter;
- the V-HOLD valve;
- the V-IN valve.

Airtuckers control:

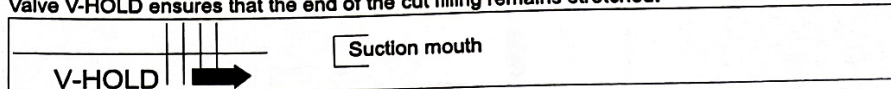
The... -print	controls ...
ATB_X	• ATI-LHS (LHS airtucker); • ATI-RHS (RHS airtucker).
ATB_Y	• ATI-MID1 (first center airtucker); • ATI-MID2 (second center airtucker).
ATB_Z	• ATI-MID3 (third center airtucker); • ATI-MID4 (fourth center airtucker).

Operation of airtucker

1. The ATB print controls the cutter so the filling is cut to the correct length.

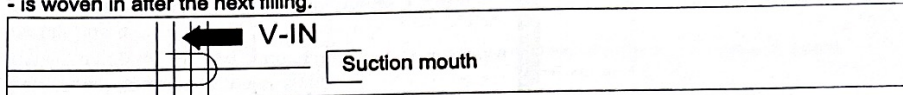


2. Valve V-HOLD ensures that the end of the cut filling remains stretched.



3. Valve V-IN ensures that the cut filling:

- is blown into the shed;
- is woven in after the next filling.

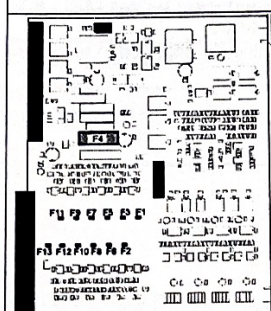


Signals

Fusing

There are no LEDs on the print.

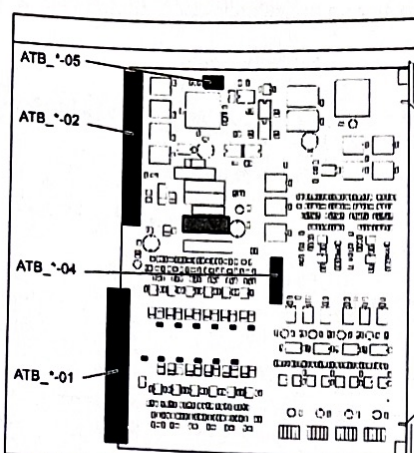
Illustration



Fuse	ATB_X	ATB_Y	ATB_Z	Value	Comment
F1	AT-V-HOLD-RHS	AT-V-HOLD-MID1	AT-V-HOLD-MID3	1,6 AT	soldered onto the print.
F2	spare				
F3	AT-V-IN-RHS	AT-V-IN-MID1	AT-V-IN-MID3	4 AF	—
F4	RHS and LHS airtucker cutter motor	MID1 and MID2 airtucker cutter motor	MID3 and MID4 airtucker cutter motor		
F5 up to and incl. F7	spare			1,6 AT	soldered onto the print.
F8	AT-V-IN-LHS	AT-V-IN-MID2	AT-V-IN-MID4		
F9	spare				
F10	AT-V-HOLD-LHS	AT-V-HOLD-MID2	AT-V-HOLD-MID4		
F11 up to and incl. F13	spare				

Connections

Illustration



Connector	ATB_X	ATB_Y	ATB_Z
ATB-01	- RHS and LHS airtucker cutter motor - airtucker valves LHS and RHS	- MID1 and MID2 airtucker cutter motor - airtucker valves MID1 and MID2	- MID3 and MID4 airtucker cutter motor - airtucker valves MID3 and MID4
ATB-02	directly plugged in to SBPO-04	directly plugged in to SBPO-05	directly plugged in to SBPO-06
ATB-03	not present on the print		
ATB-04	for Picanol service		
ATB-05			

TRIAX control box (CB-TRIAX)

Contents

- General [81]
- Location [81]
- T_TRIAX print [81]

General

Electrostatic discharges

The printed circuit boards contain integrated circuits that are sensitive to electrostatic discharges:

- Put the printed circuit boards away in special boxes.
- Do not place the printed circuit boards on a metal surface, to avoid short-circuiting the batteries on the board.
- Do not touch the printed circuit boards unnecessarily and handle them with care to avoid permanent damage to the circuits.



Switch off the mains before opening the door of the TRIAX main control box.

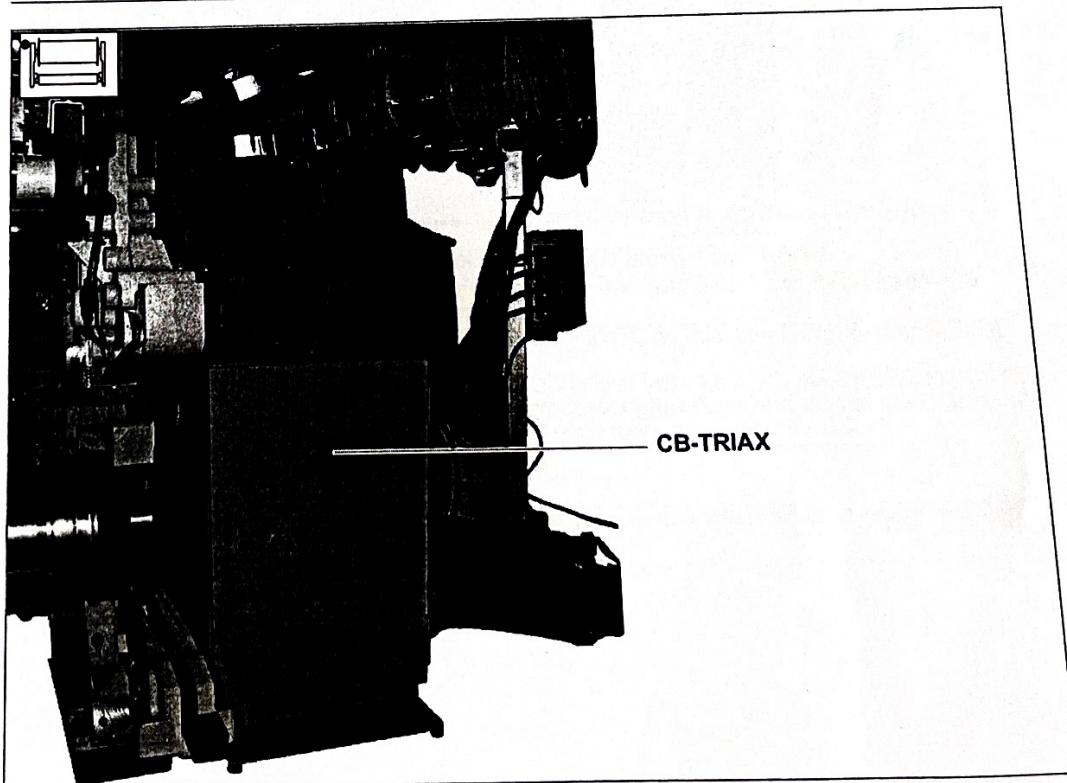
Power supply

Power to the TRIAX control box is supplied from the main control box.

Voltages

Voltage	Description
+290V DC	power supply to the SR motors
+26VDC1	brake and bimetal in the SR motors
-12V DC	logic voltages on the T_TRIAX print
+12V DC	
+5V DC	

Location



T_TRIAX print

Name

TRI AXes

Version

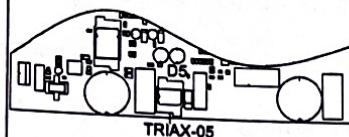
Version	Machine version	This print controls...	Can be replaced by ...
TRIAX/D-2: BE303188	double beam, with Optileno	ELO2 (LHS)	• TRIAX/D-2 • TRIAX-2
	fancy let-off on machine, with Optileno	ELO-FA	

Safety

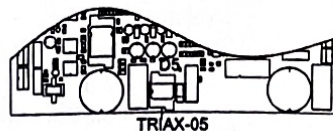
Before working on the print:

1. Set main switch QM to position -0-.
2. Wait for the green LED D5 to fade completely, because the capacitors discharge only slowly.

T_TRIAX/D-2



T_TRIAX-2

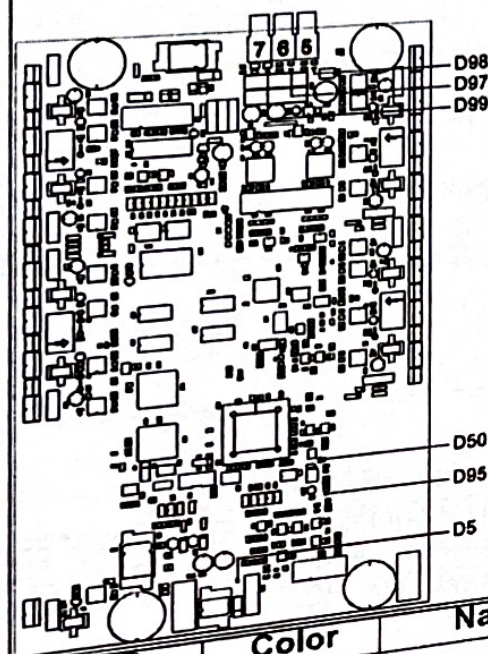


- This print contains a microprocessor using its own software. This software is checked upon:
 - power-on;
 - each change of machine software.
- If the check indicates that the software is not compatible with the MCB print software, new software is automatically loaded into the T_TRIAX print.
- This is an air-cooled print. The cooling system of this print can be severely disrupted by dirt such as dust (dressing). As a consequence, to ensure the proper operation of the T_TRIAX print it is very important that the air circulation on and around the cooling plate of the print is kept optimal. The heatsink of the print should be blown clean regularly (with every gaiting).

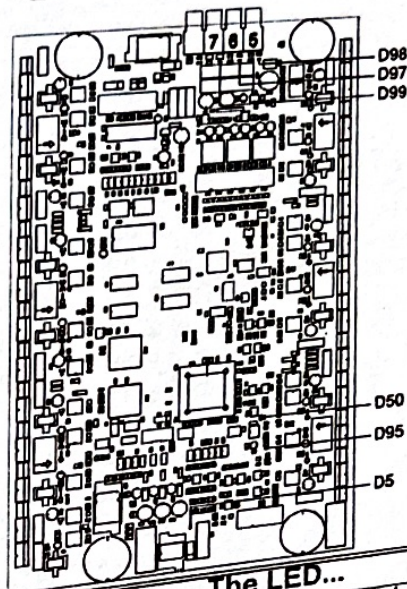
The T_TRIAX print controls the ELO2 motor or the ELO-FA motor on the basis of the warp tension of the driven warp beam.



T_TRIAX/D-2

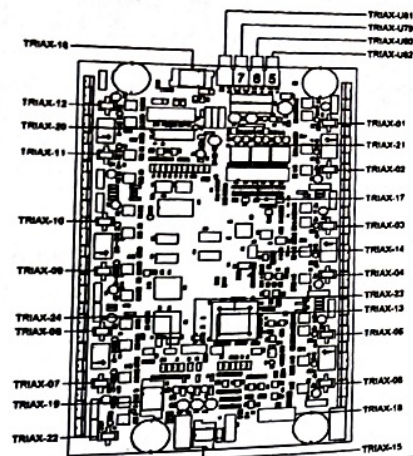


T_TRIAX-2



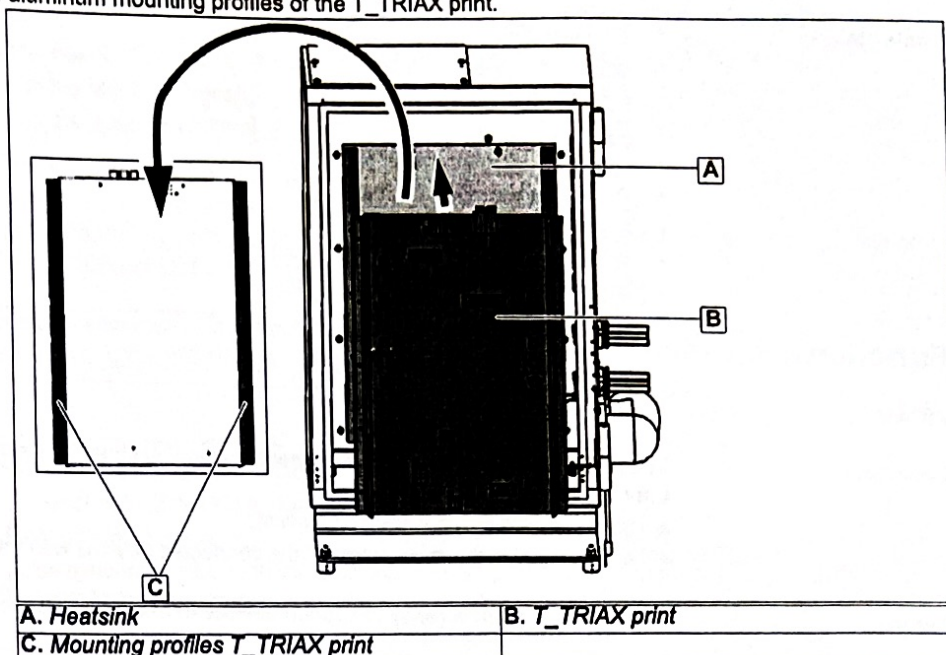
LED		Color	Name	The LED...
D5	green	green	—	lights up if +290V is present on the print.
D50			Watchdog	flashes fast if the microprocessor and the software are operating.
D95			Reset	lights up if the +5VDC on this print is higher than 4,8VDC.
D97	yellow	yellow	HS-CAN-IN	If the +5 VDC is less than 4,8V DC, the RESET line is low.
D98			HS-CAN-OUT	The microprocessor goes down and will not start again until the +5 VDC once more exceeds 4,8V DC.
D99			RS-IN	is not used.
				flashes while the optical angle position is being received.

There are no fuses on the print.

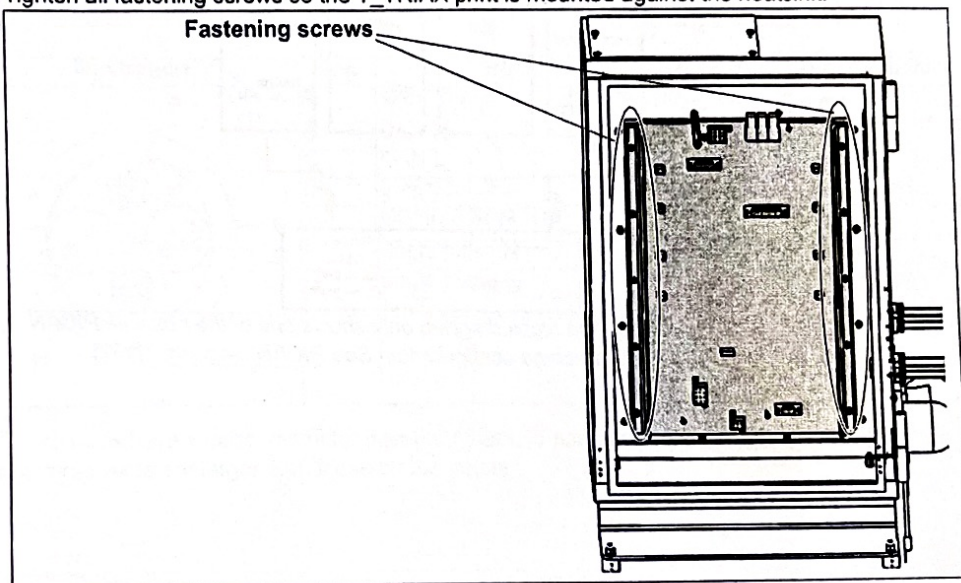


Connector	Connections
T_TRIAX-01	Phase A
T_TRIAX-02	Phase C
T_TRIAX-03	Phase B
T_TRIAX-04	Phase D
T_TRIAX-05	Phase A
T_TRIAX-06	Phase C
T_TRIAX-07	Phase B
T_TRIAX-08	Phase D
T_TRIAX-09	Phase A
T_TRIAX-10	Phase C
T_TRIAX-11	Phase B
T_TRIAX-12	Phase D
T_TRIAX-13	For Picanol service.
T_TRIAX-14	Power supply coming from the EPDAF print.
T_TRIAX-15	Power supply and DD-2 coming from the EPDAF print.
T_TRIAX-16	Signals from the resolvers: Left-hand ELO motor or fancy beam ELO motor.
T_TRIAX-17	CAN communication with the MCB print.
T_TRIAX-18	- Power supply; - Bi-metals and brakes: Left-hand ELO motor or fancy beam ELO motor.
T_TRIAX-19	Right-hand waste cutter (with electrical waste cutters).
T_TRIAX-20	For Picanol service.
T_TRIAX-21	Dump resistor.
T_TRIAX-22	Not used.
T_TRIAX-23	DATA IN.
T_TRIAX-24	CAN bus (OUT).
U79 (7)	Spare output (optical).
U80 (6)	CAN bus (IN).
U81	
U82 (5)	

-  When replacing the T_TRIAX print, apply some cooling paste between the heatsink and the 2 aluminum mounting profiles of the T_TRIAX print.



-  Tighten all fastening screws so the T_TRIAX print is mounted against the heatsink.



PICAN bus

Contents

- [Function and description](#) [86]
- [Signals and cable](#) [87]
- [EPDAF print](#) [87]
- [CJB print](#) [89]
- [CAN and PICAN devices](#) [89]
- [PICAN devices > Prewinders](#) [90]
- [PICAN devices > PFL](#) [92]
- [PICAN devices > Thermal cutter \(eCTTH\)](#) [95]
- [PICAN devices > Jacquardette](#) [96]

Function and description

PICAN

Picanol Controller Area Network

Description

The PICAN system is a bus system based on the CAN bus. This simple and flexible system enables the machine to:

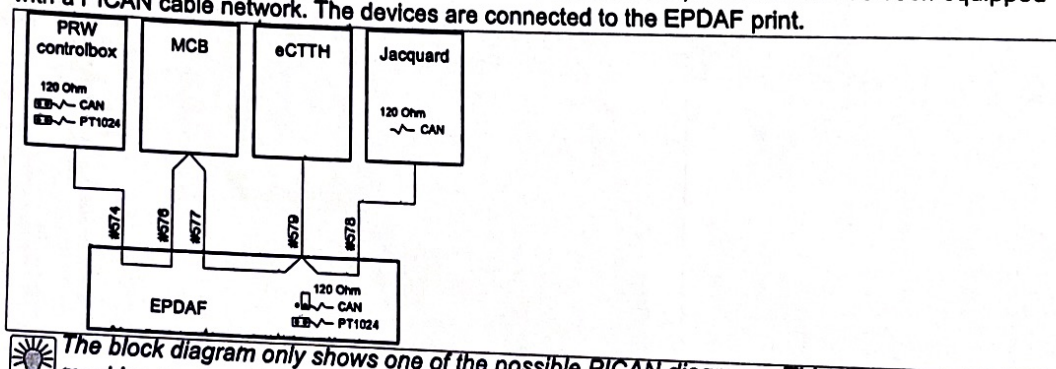
- communicate with connected devices;
- to synchronize the movements of the connected devices with the machine movements.

Future

The PICAN-System is ready to support devices in future that do not yet exist today.

General

PICAN devices are connected to the PICAN bus. To allow for this, the machine has been equipped with a PICAN cable network. The devices are connected to the EPDAF print.



 The block diagram only shows one of the possible PICAN diagrams. This depends on the machine configuration. See [PICAN diagram](#) [132].

"Termination"

The "Termination" is necessary to prevent signal distortion.

The machines always come supplied with "Termination" at both ends. These terminations are necessary for the proper operation, even if no PICAN devices are present on the machine.

The "Termination" comprises two 120 Ω terminal resistors:

- 1 on the MS-CAN signal;
- 1 on the pulse train signal.

The "Termination" is situated:

- on the EPDAF print (machine without jacquard),
or on the EPDAF print (pulse train) and Jacquard controller (MS-CAN) (machine with jacquard);
- in the IRO controlbox.

If problems occur

If problems occur with the Pican bus, the resistance of the Pican bus must be measured.
See Measuring the PICAN bus [88].

Signals and cable

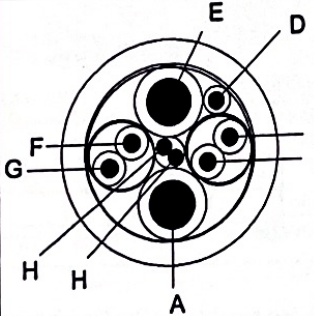
Signals

The PICAN bus carries the following signals:

- power supply (26V DC) for local controls;
- CAN communication;
- pulse synchronization with the machine;
- EVENTLINE for exceptions.

These signals are bundled in one flexible cable which is connected by means of an 8-pole connector (round plug).

Cable

Illustration	Peg	Signal	Color	Wire diameter
	A	+26V DC	red	2 mm ² (1)
	B	CAN H	white	0,34 mm ²
	C	CAN L	brown	
	D	EVENTLINE	grey	
	E	CGND (-26V DC)	black	2 mm ² (1)
	F	angle pulse H	green	0,34 mm ²
	G	angle pulse L	yellow	
	H	cover	—	

(1): These wires have a larger diameter than the others, to prevent voltage drop. Accordingly, the pins of these wires are larger than those on the others.

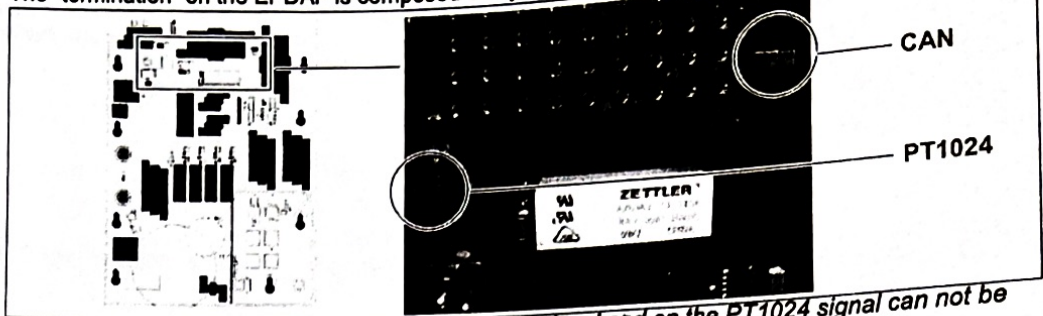
EPDAF print

General

- The EPDAF print connects all PICAN signals to the PICAN devices.
For more information on the PICAN principle, see: Function and description [86].
- For more information on the EPDAF print, see: EPDAF/R print [38].

Termination on the EPDAF

The "termination" on the EPDAF is composed of 2 jumpers.



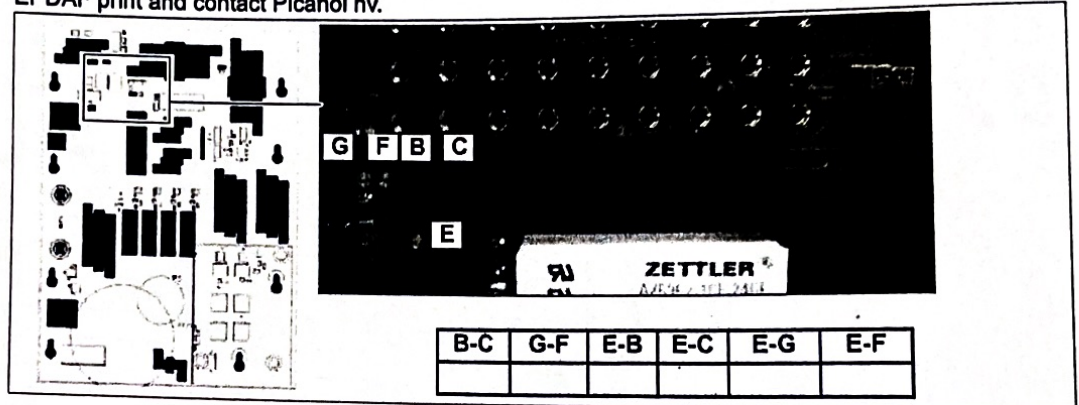
In some cases, the termination on the MS-CAN signal and on the PT1024 signal can not be found on the EPDAF print.

The EPDAF Pican termination with ...	CAN	PT1024	The termination can be found in the:
standard version	1	1	/
Jacquard machine	0	1	Jacquard
Fancy behind the machine	0	0	WTB print

1: jumper installed
0: jumper not installed

Measuring the Pican bus

In case of problems with the Pican bus: Measure the resistance between the following points on the EPDAF print and contact Picanol nv.



Replacement of the print

When replacing the EPDAF print, the position of the termination jumpers must be checked, see: Termination on the EPDAF [88].

Signals

Every LED color has a specific meaning:

- Green LED: Indication.
- Yellow LED: danger, don't touch (i.e.: at high voltages)
- Red LED: Indicates a problem (e.g. a blown fuse)

LED	Color	The LED is lit if...	
D3	green	48V AC	is present.
D25	yellow	+26V DC	
D32		+290V DC	
D17	red	+26VDCPICAN	is not present.
D19		+26VDCS	
D21		+26VDC1	
D23		+26VDC2	

CJB print

Name

Can Junction Box

Version

Print: CJB-1: BE235119

Function

The PICANOL-PFL and the prewinders are connected to this print.
 The CJB print is only present if the machine is equipped with PICANOL-PFL.

Signals and fusing

There are no LEDs or fuses on the print.

Connections

Illustration	Connector	Connections
	CJB-01	PICAN signals coming from the EPDAF print
	CJB-02	Connection for PICAN devices.
	CJB-03	• prewinders controlbox; • PICANOL-PFL.

CAN and PICAN devices

CAN <--> PICAN

With CAN devices the angle position is not passed on through the PICAN cable, but in another way (e.g. by means of a mechanical shaft).

Switch off the mains before (dis)connecting the (PI)CAN devices to/from the machine.
The (PI)CAN device is detected after the mains is switched on again.

CAN devices

The CAN devices are:

- jacquard;
- CAN hobby